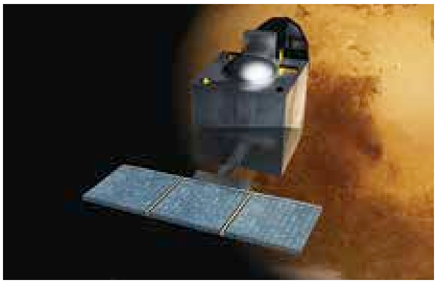

The highly successful Mangalyaan mission was the cheapest Mars orbital mission till date. ISRO would be looking to repeat the success story with its next mars mission

another orbiter. The Mangalyaan 2 will be built as a joint venture between India and France, and a letter of intent has already been signed by ISRO and CNES. The Mangalyaan 2 will be completed by the year 2020 and will use GSLV III, the same heavy lift launcher that will be used for the Chandrayaan.

- **Venus Orbiter Mission** - A proposed mission to study the atmosphere of Venus, the Venus orbiter mission was slated for launch somewhere between 2017-2020, if funded. There hasn't been any major updates regarding the mission, but it is safe to assume that India will keep on increasing its interplanetary exploration capabilities and will follow up the Chandrayaan-2 and Mangalyaan-2 mission with a trip to Venus.
- **Solar Exploration Programme**- NASA and ESA are the only two space agencies to have successfully placed a satellite at the 'Langarian Point L1' near the Sun. India's solar mission aims to change that with Aditya-L1. The probe, which weighs around 400 kgs, was originally set for a launch in the year 2012 but due to complex and extensive work required in fabricating a satellite capable of standing the extremities of outer space at such a close distance to the Sun, the launch was postponed to 2020. The main aim of Aditya 1 is to study Coronal Mass Ejection and the Coronal magnetic fields which might have a significant impact on the unsolved problem of heating of the corona (the upper atmosphere of the sun is a million degrees celsius hot, while the lower atmosphere's temperature measures at 6000° C) that has been bothering scientists for a long time now. The spacecraft will be carrying more than 7 payloads, each consisting of various scientific instruments to measure quite a few things including the variation and composition of solar winds
- **Space capsule recovery experiment II** - A follow up to the successful Space Capsule Recovery-1 mission which was successfully completed in the year 2007, the SRE-2 aims to develop a fully recoverable capsule, and perform a number of experiments on Microbiology, Powder Metallurgy and agriculture in microgravity. The spacecraft will also be carrying an isothermal furnace capable of handling temperatures upto 1000°C in outer space.
- **Manned Space programme** - Sending people out in the space isn't as easy as they make it look like in your favorite sci-fi movie. Sending humans out in the space requires significantly advanced technology and billions of dollars in budget. Currently ISRO doesn't have any human rated launch vehicle but the race is on. ISRO is currently working on a 3 ton orbital vehicle spaceship that is capable of carrying a 2 member crew to space and return after a couple of days. ISRO plans to set up a training centre in Bangalore specifically to train Indian Astronauts or 'Vyomanauts' (vyoma means space) and also is building a third launch pad at the Satish Dawan Space centre specifically for launching manned space vehicles. But the road is a long and tough one and there haven't been many updates from ISRO regarding the expected date of launch. Though, if all goes according to plan we will see an Indian in space somewhere in the next decade, making India only the 4th nation in the world, after China, USSR and USA, to successfully carry out a manned mission indigenously.

Rakesh Sharma and Kalpana Chawla are the only 2 Indian astronauts to have gone out into space

CONCLUSION

Even though ISRO is one of the youngest space organisations, it's achievements are highly commendable and has earned it global repute. It has many interesting projects and innovative ideas, and turning them into reality requires a lot of hard work, time and money. ISRO can't be compared to NASA and other such organisations, partly because of the head start they have had and partly, because their aim is different. While NASA, European Space Agency and Russian Space Agency work on a variety of domains, ISRO's main focus is on creating various utilitarian technologies like weather and geographic satellites and launching probes. ISRO has come a long way and it's a long road ahead. ISRO has a pool of some of the smartest and most dedicated individuals in the world. Who knows, maybe "Indian space agency finds life in our solar system" is the next headline we read.